

Reading Support: Describing Admissible Strings I

Companion to Week 6

Cem Bozsahin

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A formal grammar names four components

Supports: A grammar is a finite description of a stringset

A grammar $G = (V, \Sigma, R, S)$ has nonterminals V , terminals Σ , rewrite rules R , and a distinguished start symbol S . The sets V and Σ are finite and disjoint.

Its language is $\mathcal{L}(G) = \{w \in \Sigma^ \mid S \xRightarrow{*} w\}$: the terminal strings derivable from the start symbol in zero or more licensed steps.*

MOLR §4

Restrictions on rules define grammar families

Supports: Rule form determines descriptive power

A context-free rule rewrites one nonterminal independently of its surrounding string. A regular rule permits still less: the continuing nonterminal, when present, occurs at one edge of the right-hand side. Context-sensitive and unrestricted grammars allow more material on the left.

Changing rule form changes the possible descriptions even before any particular derivation is attempted.

MOLR §4

Derivability closes one-step rewriting under repetition

Supports: Derivation is a relation over intermediate strings

If a rule replaces one occurrence of its left-hand side, the resulting pair of strings belongs to the one-step derivation relation $\Rightarrow$. Its reflexive transitive closure $\Rightarrow^$ permits zero or more steps.*

A derivation records one route from S to a terminal string. The language includes the terminal endpoint, not the incidental order in which independent rewrites happened.

MOLR §4

A context-free grammar can coordinate paired growth

Supports: One continuing nonterminal can coordinate two edges

The rules $A \rightarrow aAb \mid ab$ generate equal positive numbers of a 's and b 's. Each recursive step contributes one terminal at either edge of the material derived from A .

With $S \rightarrow AB$ and a parallel grammar for $c^m d^m$, the two dependencies remain independent constituents inside one generated string.

MOLR §4.1

Stringset identity leaves structural questions open

Supports: The same stringset can receive different structures

Weakly equivalent grammars generate the same terminal strings. Strong equivalence requires an appropriate correspondence among their structural descriptions as well.

Linguistic analyses ordinarily make claims about dependency, constituency, and interpretation. Weak equivalence can therefore preserve successful recognition while losing the structure that motivated the grammar.

MOLR §4.1

A derivation tree suppresses irrelevant rewrite order

Supports: A derivation tree is not merely a record of rewrite order

A derivation tree places the start symbol at its root, rule daughters under each rewritten nonterminal, and terminals along the yield. Independent branches need not remember which branch was rewritten first.

The tree preserves dominance and constituency. More than one tree for the same yield constitutes structural ambiguity when the difference matters to the analysis.

MOLR §4.1

Linear indexes add a controlled stack to one path

Supports: Indexes pass a controlled dependency through a derivation

In a linear-indexed grammar, nonterminals can carry a stack of index symbols. Rules add, pass, or remove indexes, but only one daughter continues the indexed path.

This restricted memory supports dependencies beyond ordinary context-freeness while preventing arbitrary access to the stored information throughout the derivation.

MOLR §4.2

Mild context-sensitivity names a constrained extension

Supports: Mild context-sensitivity targets limited cross-serial dependency

Some syntactic dependencies motivate descriptions stronger than context-free grammars. Natural-language evidence does not thereby justify unrestricted rewriting.

Linear-indexed and related formalisms aim to capture limited cross-serial or multiple dependencies while retaining constrained growth and feasible recognition. The empirical burden concerns the exact dependency, not the label attached to a whole language.

MOLR §4.2